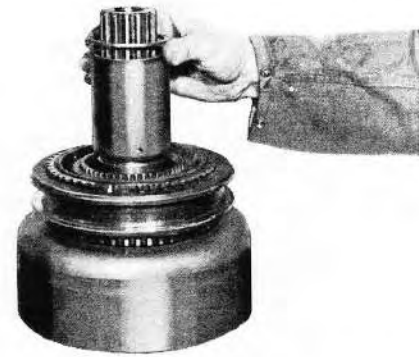


717.900
SA 35 977

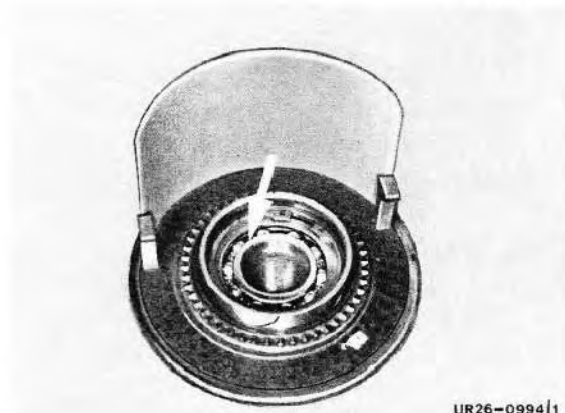
- 11 Fit distance tube and compensating shim.

Note: Fit oilway downwards.



UR 26-0828

- 12 Fit deep groove ball bearing with compensating shim and secure, coat oil deflector plate with adhesive No. 4 and fit. See 1.5/1 for perm. end play.



UR26-0994/1

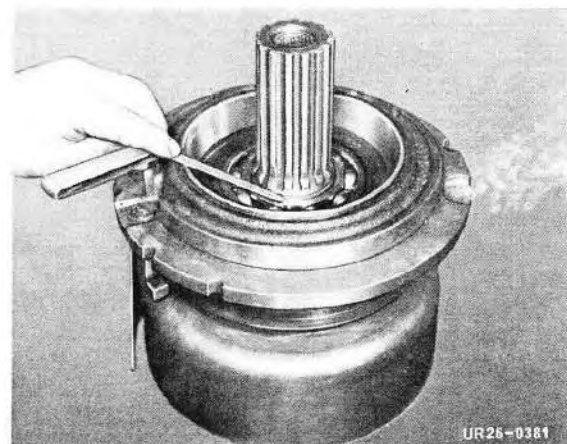
- 13 Fit synchro hub.



UR 26-0804/1

- 14 Measure synchro hub and secure.
Permissible end float, see 1.5/1.

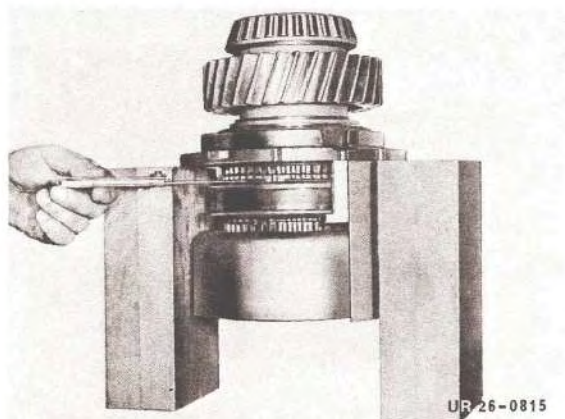
- 15 Place input gear on planet carrier and fasten with bolt and spacer.



UR26-0361

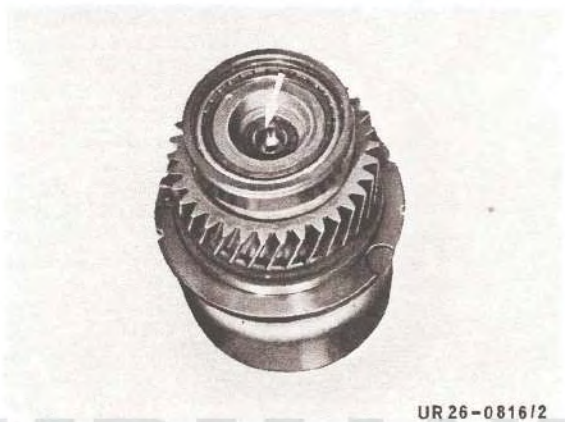
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16 Support planet carrier and check the synchromesh play.

Note: Should the permissible synchromesh play not be reached, the epicyclic gear train must be dismantled and the operations 11 to 16 repeated.



17 If an auxiliary transmission is fitted, unscrew the bolt, measure the bearing cap and input gear and determine the compensating shims.

Permissible end float, see 1.5/1.

18 Remove input gear and fit selector fork.

19 Install planetary gearing.

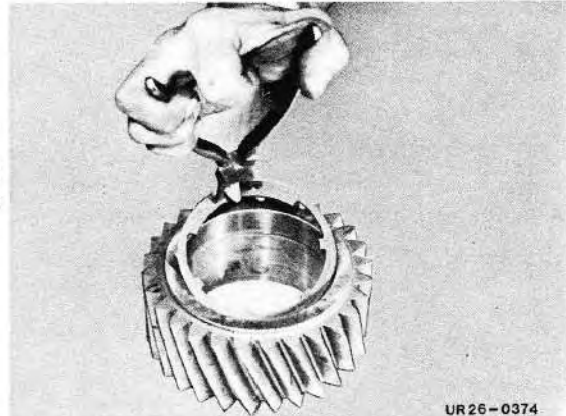
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Assembly of Transfer Case

To Transmission No. 008 096

Note: The taper roller bearings on the idler gear shaft are not measured since the bearings and spacing sleeves should be obtained only as a repair kit.

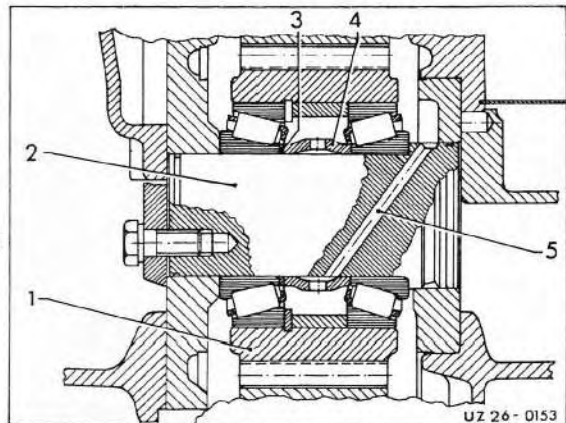
- 1 Install retaining ring and spacer sleeve.



UR 26-0374

- 2 Press outer race of taper roller bearing.
- 3 Press inner races of taper roller bearing together with spacing sleeve (3) and shim (4) on to the idler gear shaft.
- 4 Install idler gear (1) and idler gear shaft (2) in transfer case.

Note: Observe the position of the oil duct (5) and sealing rings on the idler gear shaft.



UZ 26-0153

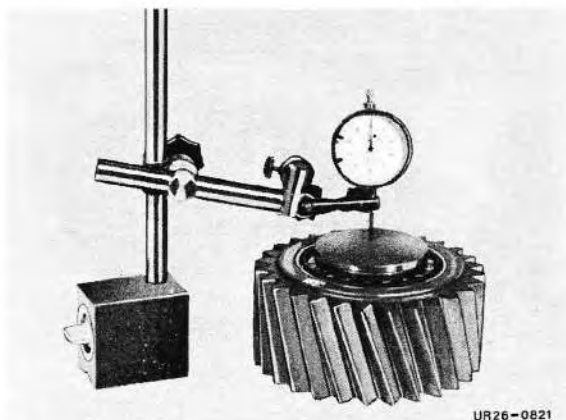
From Transmission No. 008 097 to Transmission No. 012 814

- 5 Install both taper roller bearings in idler gear.



UR 26-082911

- 6 Attach special tools No. 11, No. 17 and No. 18, turn idler gear and bearings several times, adjusting special tool No. 18 to "0".



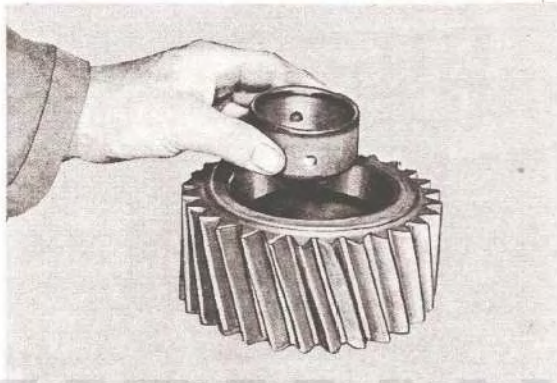
UR 26-0821

26.13 Assembly of Main Transmission

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UR 26-0829/1



UR26-0822

Example

$$\begin{array}{r} a \\ -b \\ \hline \\ +c \\ \hline x= \end{array} \begin{array}{l} 45,35 \text{ mm} \\ -0,21 \text{ mm} \\ \hline 45,14 \text{ mm} \\ \\ 0,06 \text{ mm} \\ \hline 45,20 \text{ mm} \end{array}$$

7 Carefully move special tool No. 17 to one side, removing special tool No. 11 and inner race of taper roller bearing.

8 Measure one spacer sleeve (a) and attach together with inner race of taper roller bearing and special tool No. 11.

9 Repeat step No. 2 and read off new measurement (b).

10 Calculate dimension of spacer sleeve X.

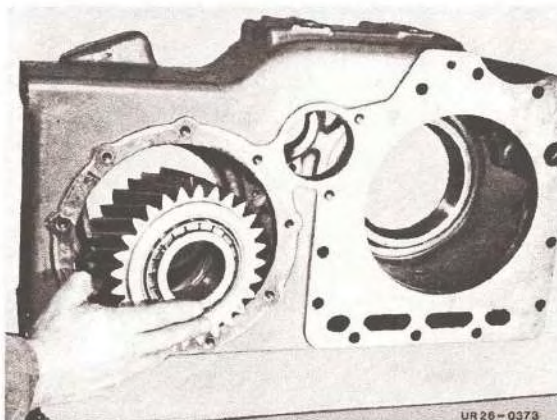
$$x = a - b + c$$

x = necessary space in mm

a = mm thickness of inserted spacer sleeve

b = reading in mm

c = permissible bearing play in mm



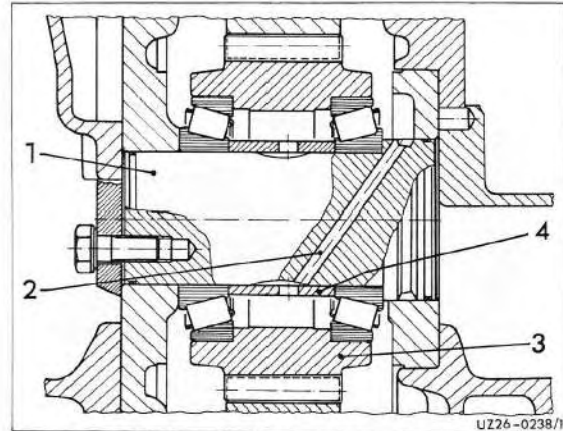
UR 26-0373

11 Install spacer sleeve and recheck calculated size as described in step 2.

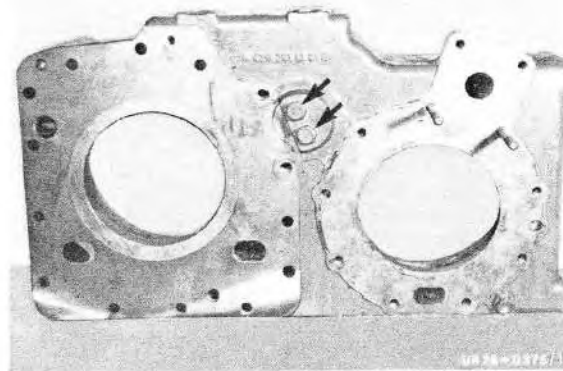
12 Install assembled idler gear.

13 Install intermediate shaft (1).

Note: Pay attention to oil duct (2) and sealing rings.
(3) Idler gear
(4) Spacing sleeve



14 Install idler gear with supporting disk and tighten. See 1.7/1 for tightening torques.



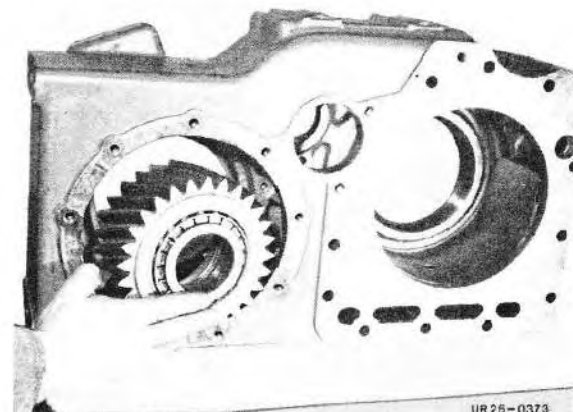
From transmission No. 012 815

Note: The spacer tube in the idler gear bearing is no longer fitted. This modification can only be exchanged in reverse in conjunction with the modified idler gear axle, the sealing rings and shims.

External identifying feature:
Shim behind supporting plate.

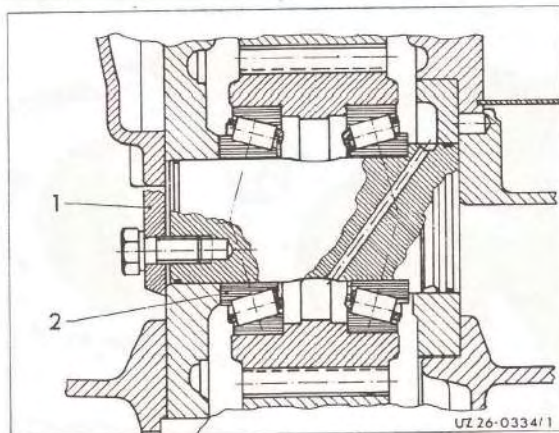
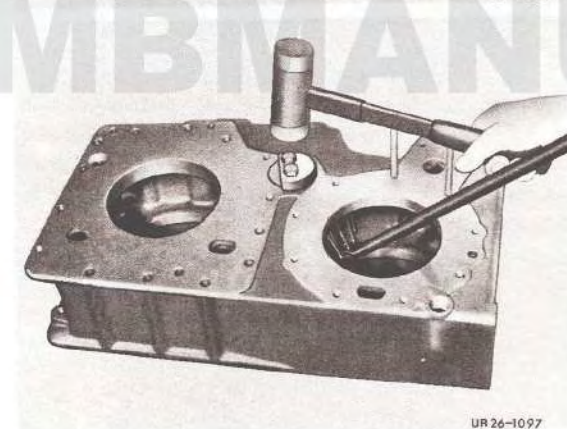
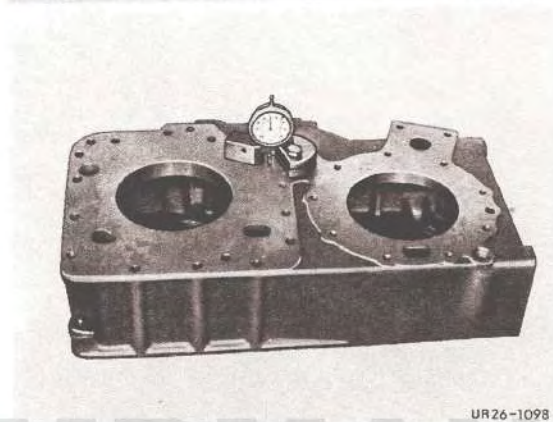
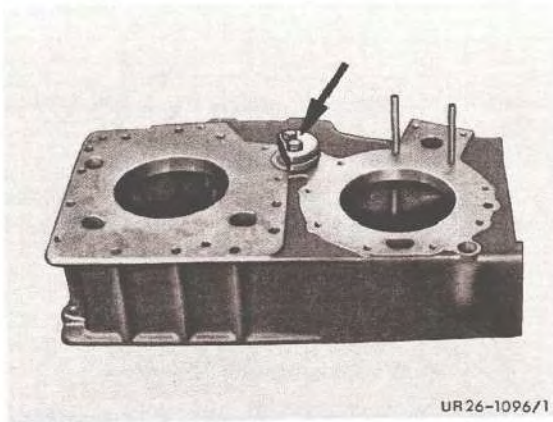
To ensure that the bearing rollers are distributed and the idler gear bearing cannot be twisted, the idler gear should be turned when tightening the bolts and prior to each measurement. Only in this way is it possible to perform a correct measurement.

1 Install idler gear with taper roller bearing in intermediate housing.



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2 Press in axle of idler gear with sealing rings.

3 Fit two supporting disks and tighten.
See 1.7/1 for perm. tightening torques.

Note: To prevent the supporting disks from being twisted, two disks must be fitted for measuring.

4 Slacken bolts and re-tighten to 30 Nm

5 Measure gap between housing and bolt with special tool No. and No. and determine the shims required. 0.25 mm must be added to the size measured.

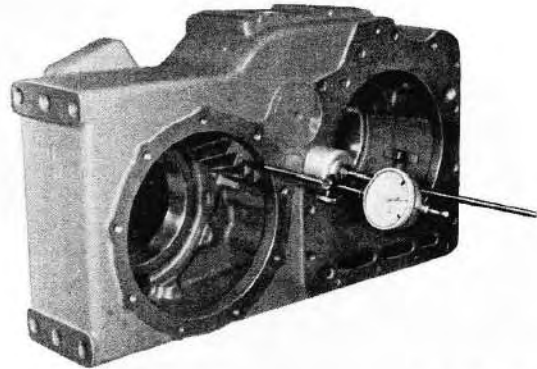
6 Remove retaining plates and place the calculated shims below. Fit retaining plate and tighten.
See 1.7/1 for perm. tightening torques.

7 Support idler gear from below with assembly lever and knock back idler gear axle.

Note: Idler gear must be raised to produce the desired end play between the roller bearings. Bearing inner race (2) and supporting disk (1) must rest against housing when the permissible end play is reached.

8 Turn intermediate housing through 90° and measure end play with special tool No. 17 and No. 18. See 1.6/1 for perm. end play.

Note: If the perm. end play is not achieved, the supporting disk must be removed and the appropriate washers removed or inserted. Then repeat No. 7 and 8.



UR26-1101

Fitting transfer box

1 Press taper roller bearing inner race onto drive gear.

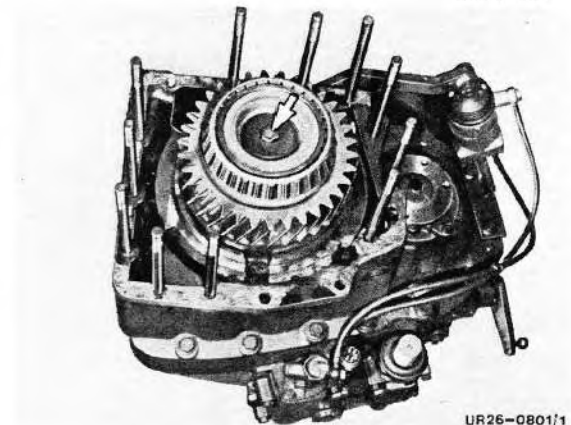
2 Install drive gear, coat bolt with locking agent No. 1, screw in and tighten. See 1.7/1 for tightening torques.

Note: From transmission No.086 019 transmission model 17.901/903 the profile gearing in the drive gear was enlarged to 72 mm.

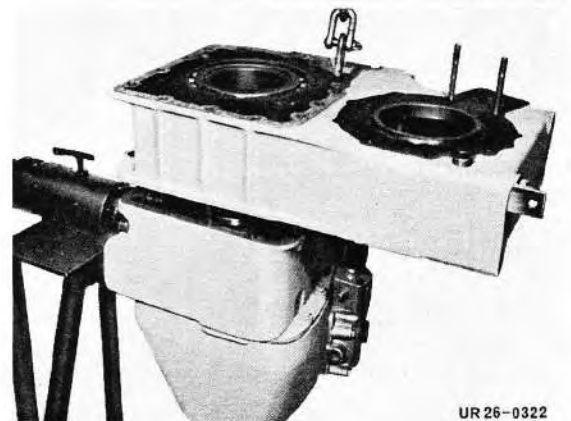
3 Fit gasket and attach transfer box casing.

4 Screw in connecting bolts and tighten. See 1.7/1 for tightening torques.

5 Press in taper roller bearing outer race.

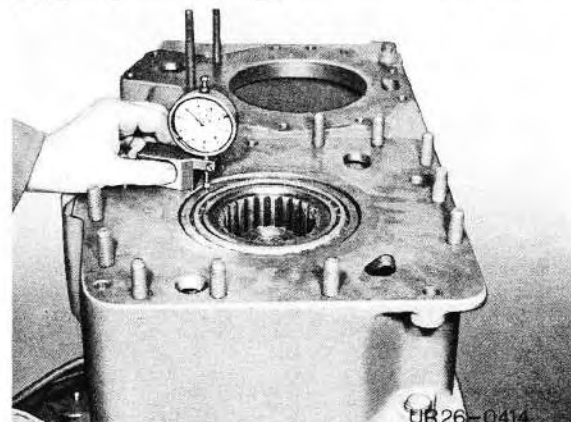


UR26-0801/1

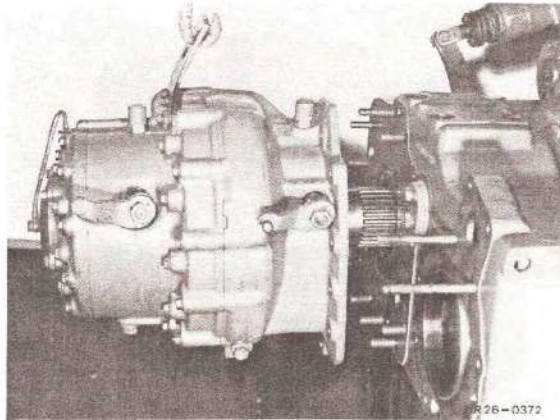


UR26-0322

6 Fit gasket and measure taper roller bearing with special tool No. 13 and No. 18, see 1.5/1 for perm. play.



UR26-0414



8 Fit bearing cover or rear auxiliary transmission.
See 1.7/1 for perm. tightening torques.

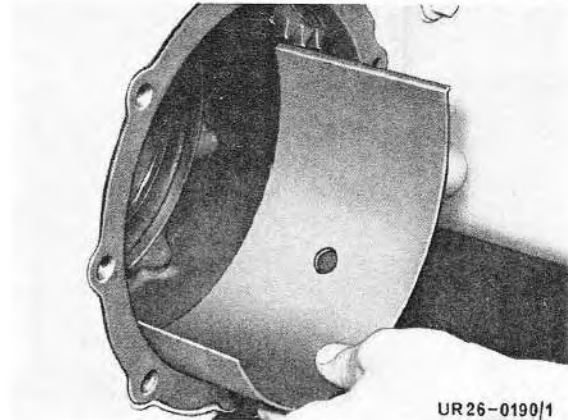
9 Fit rear axle drive, see 4.5/1.

10 Fit front axle drive, see 4.6/1.

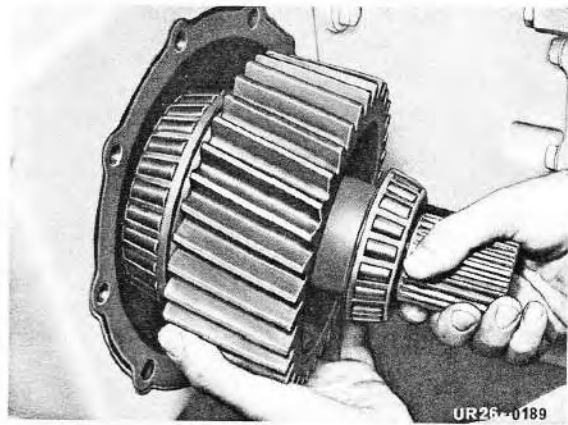
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Installation

1 Install oil baffle plate in transfer case housing.

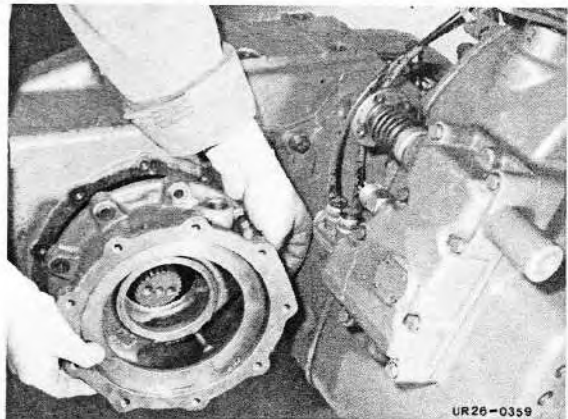


2 Press both tapered roller bearings onto drive gear and install in housing.

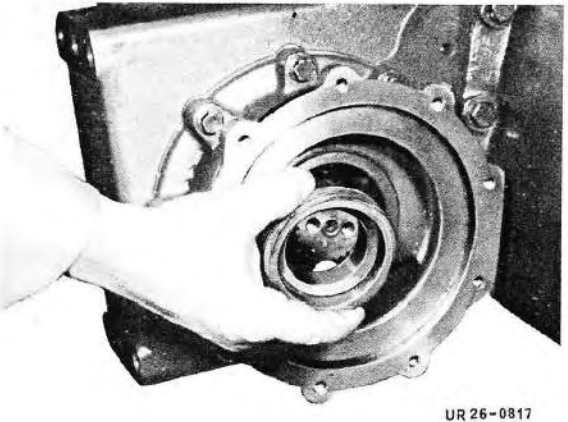


3 Install intermediate housing together with gasket so that the speedometer drive faces upwards and tighten. For permissible tightening torques, see 1.7/1.

Note: From chassis end No. 102 873, normal bolts are installed as standard in place of the locking bolts.



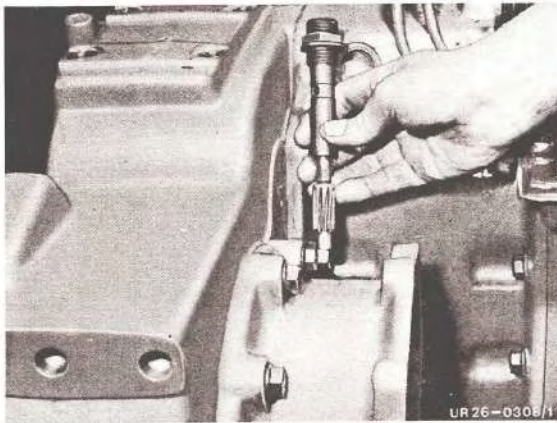
4 Install speedometer drive gear.



26.13 Installation of Rear Axle PTO

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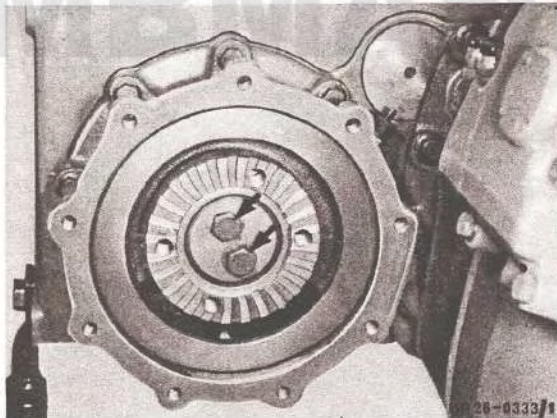
SA 35 977



5 Install speedometer shaft drive.



6 Coat shaft sealing ring with lubricating grease No. 3 and press in using special tool No. 4.

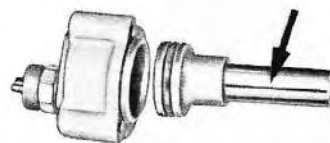


7 Place output flange in position and attach retaining plate, tighten bolts with locking washers. For permissible tightening torques, see 1.7/1.

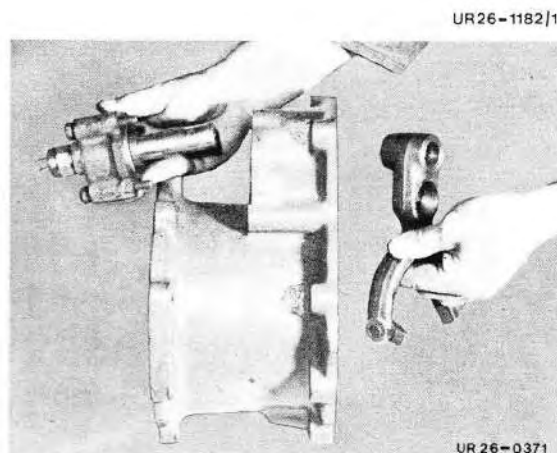
Assembly of front axle PTO

Up to transmission No. 013 286

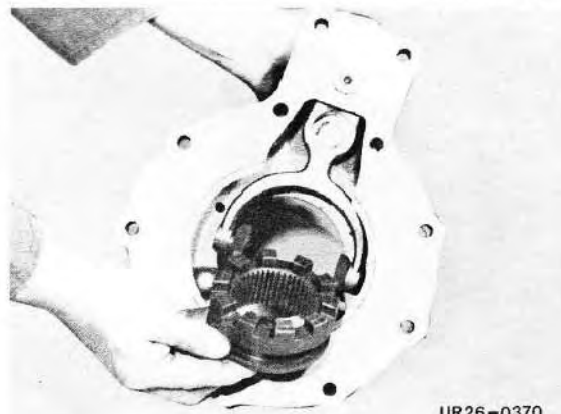
- 1 Coat shift piston with paste No. 7 and install.



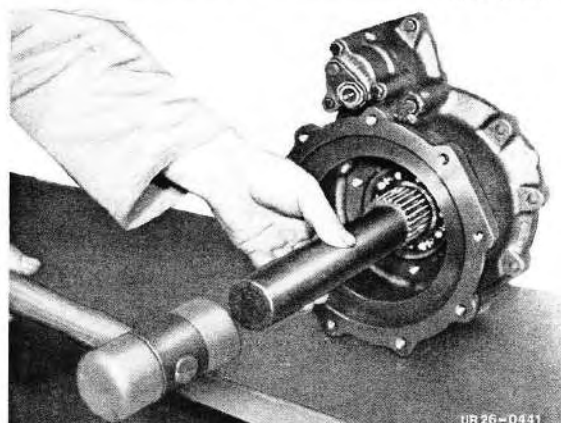
- 2 Fit shift fork with fulcrum pads for four-wheel drive and shift cylinder.



- 3 Install shift dog.
- 4 Press deep-groove ball bearing onto PTO shaft.



- 5 Install PTO shaft in intermediate housing.

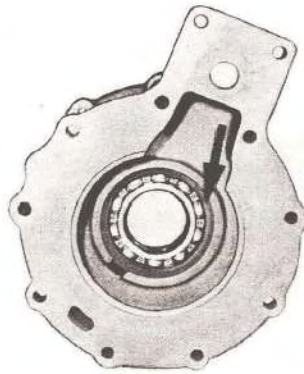


26.13 Assembly of main transmission

717.9

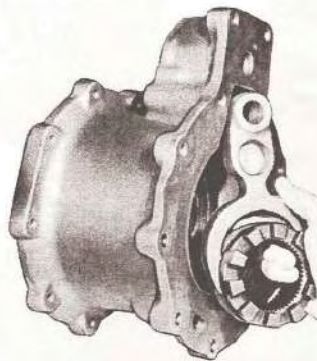
SA 35 977

From transmission No. 013 287



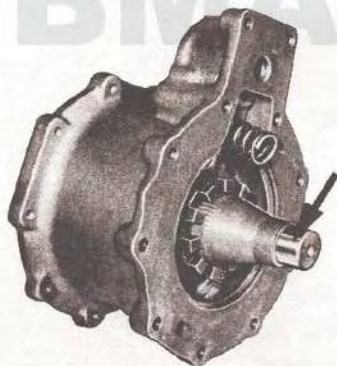
UR26-1181/1

- 6 Install deep-groove ball bearing and secure.



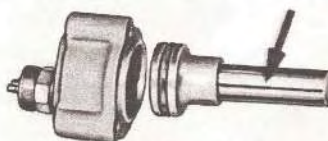
UR26-1178

- 7 Install shift dog and shift fork.



UR26-1179/1

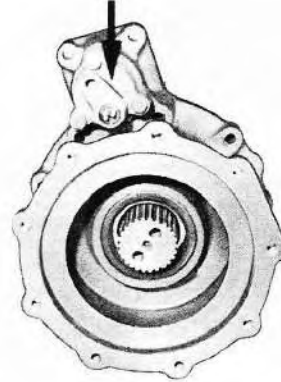
- 8 Install PTO shaft and compression spring.



UR26-1182/1

- 9 Coat shift piston with paste No. 7 and install.

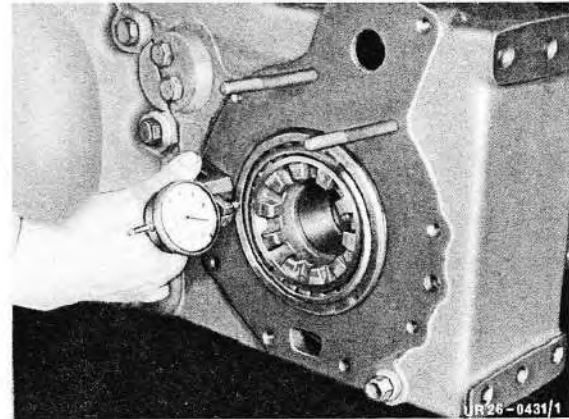
- 10 Install shift cylinder.



UR26-1180/1

Installation of Front Axle Take-Off

- 1 Check needle bearing in output gear.
- 2 Fit gasket and measure tapered roller bearing using special tool No. 13 and No. 18
For permissible axial play see 1.5/1.

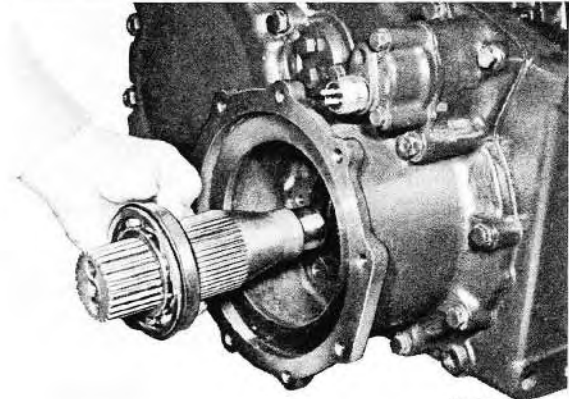


UR26-0431/1

To transmission No. 013 286

- 3 Install intermediate housing with spring, shim rings and gasket
For permissible tightening torques, see 1.7/1.

Note: From chassis end No. 102 873 normal bolts are installed as standard in place of the locking bolts.



UR 26-0818

- 4 Place gasket in position and measure bearing end cover using special tool No. 13 and No. 18
For permissible play, see 1.5/1

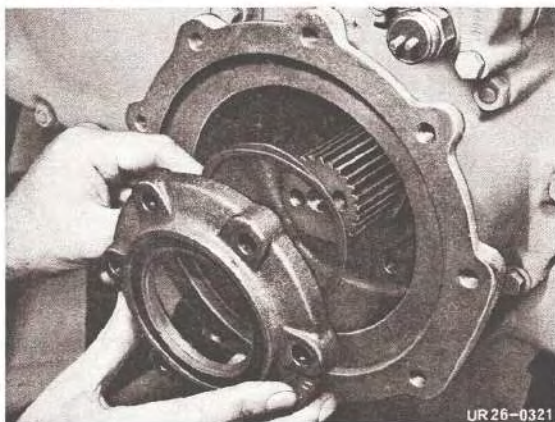


UR26-0427

26.13 Assembly of Main Transmission

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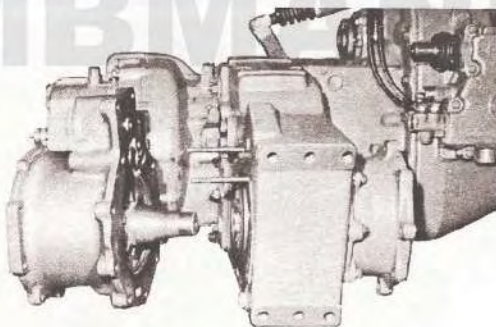
SA 35 977



5 Attach bearing end cover with gasket and shim rings, coat bolts with sealing compound No. 1 and screw in. For permissible tightening torques, see 1.7/1.



6 Coat shaft sealing ring with lubricating grease No. 3 and press in using special tool No. 4.

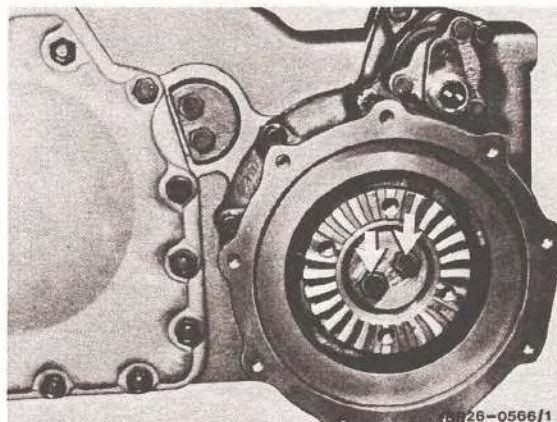


From transmission No. 013 287

7 Install intermediate housing.

Note: Do not lose shim rings and compression spring. From chassis end No. 102,873 normal bolts are installed as standard in place of the locking bolts.

UR 26-1183



To and from transmission No. 013 287

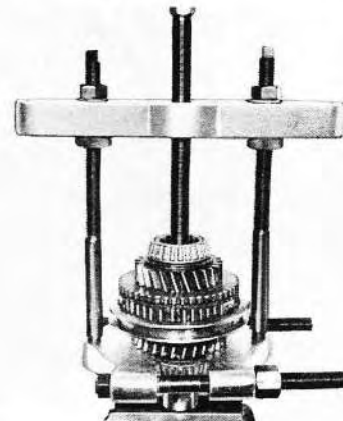
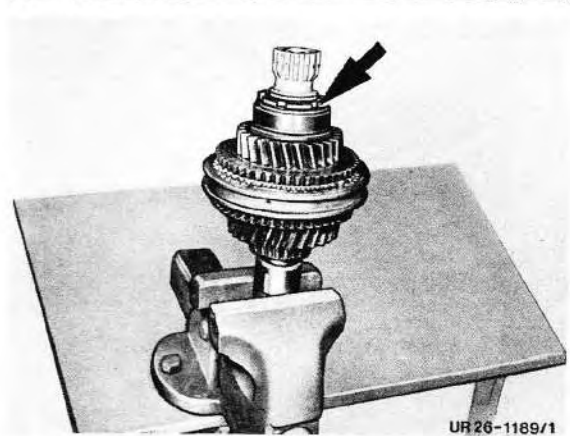
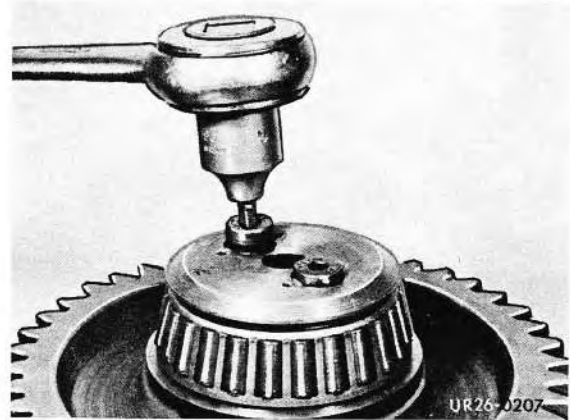
8 Mount output flange, coat retaining plate with sealing compound No. 2 and tighten with bolts and locking washers

For permissible tightening torques, see 1.7/1.

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SA 35 977

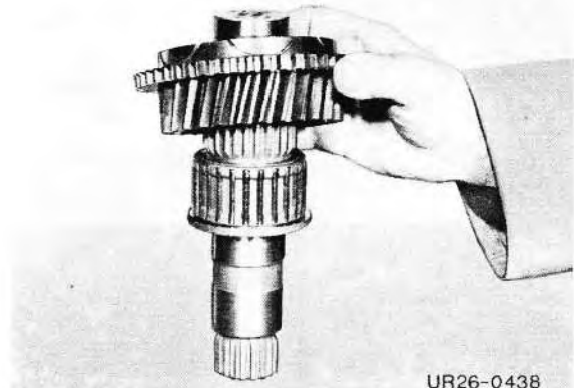
Disassembly

- 1 Clamp input shaft at drive stub shaft.
- 2 Remove retaining plates.
To Transmission No. 002 102
- 3 Unscrew slotted nut using special tool No. 7a.
From Transmission No. 020 370
- 4 Clamp input shaft and detach gear train.
- 5 Disassemble synchromesh, see 7.1/1.
- 6 Clean all parts, check and renew if necessary.



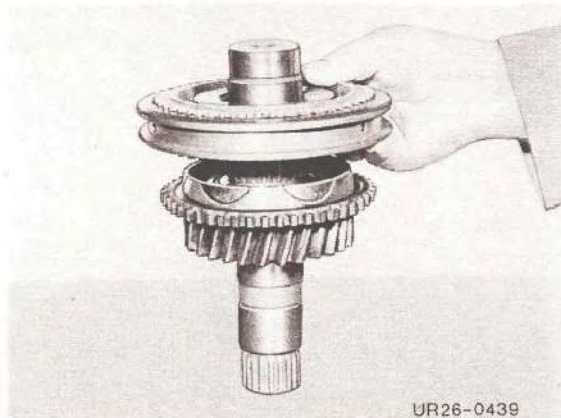
Assembly

- 1 Install needle bearing and forward speed gear.
- 2 Assemble synchromesh, see 7.1/1.

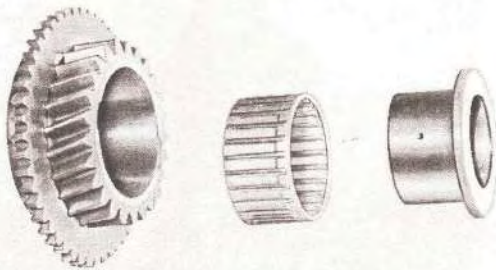


26.13 Disassembly and Assembly of Input Shaft

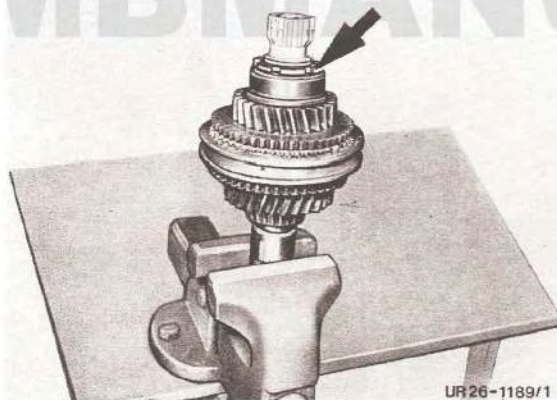
717.9
SA 35 977



3 Press synchronesh member into position.

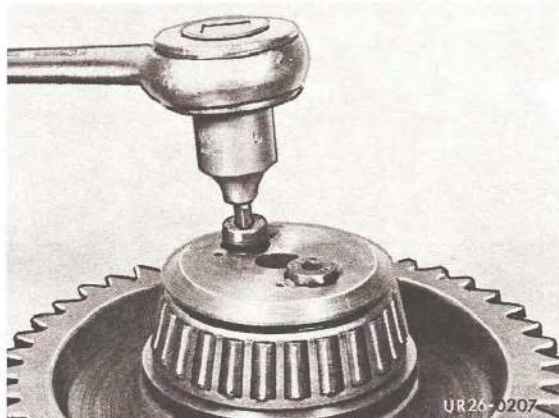


4 Install reverse speed gear, needle bearing and bearing bush.



5 Screw on slotted nut using special tool No. 7a and tighten.
Permissible tightening torques, see 1.7/1.
From transmission No. 020 370

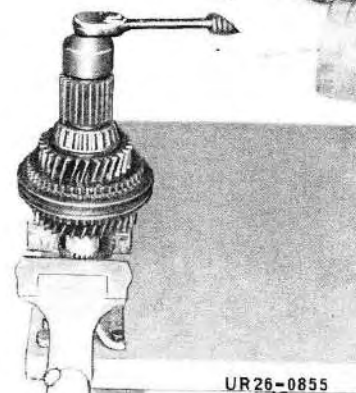
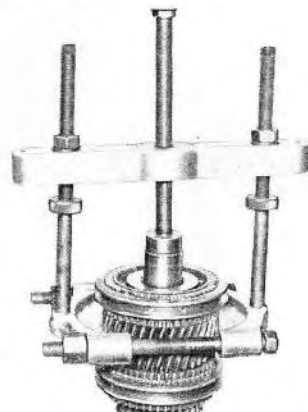
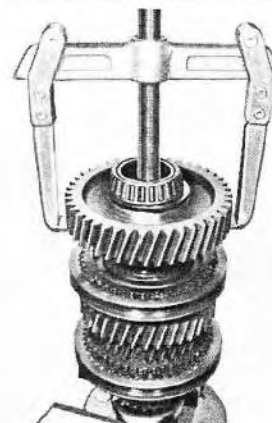
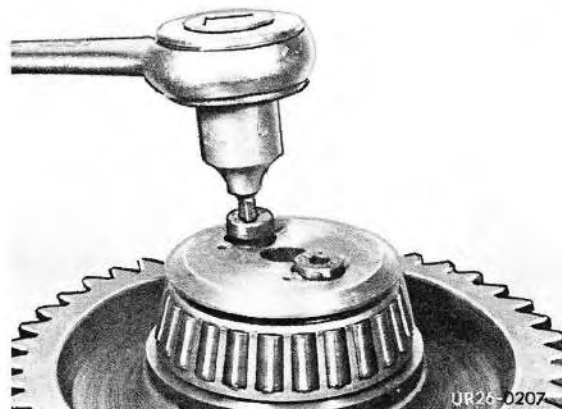
6 Press on inner race of taper roller bearing.



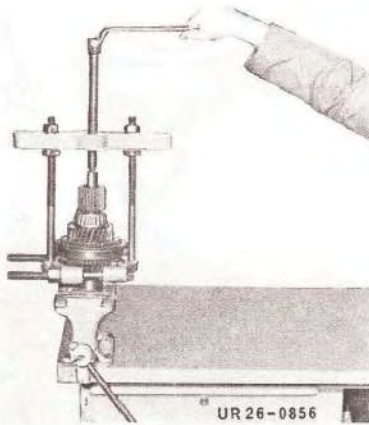
7 Screw on retaining plate and tighten.
To transmission No. 002 102.

Disassembly

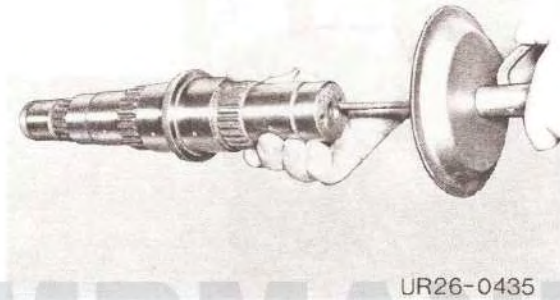
- 1 Clamp main shaft at sun gear.
- 2 Detach supporting plate.
- 3 Pull first speed gear off.
- 4 Pull second speed gear and synchromash member off.
- 5 Disassemble synchromash member, referring to 7.1/1.
- 6 Turn main shaft by 180°.
- 7 Unscrew slotted nut at sun gear yielding special tool No. 8.



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SA 35 977



- 8 Detach spacing sleeve.
- 9 Remove sun gear and 3rd/4th speed gearset.
- 10 Disassemble synchromash member, referring to 7.1/1.

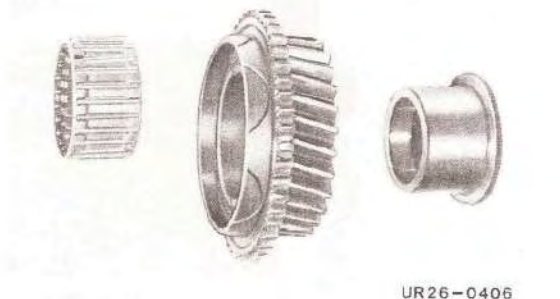


- 11 Clean oil drilling in main shaft.
- 12 Clean and check all parts, exchanging if necessary.



Assembly

- 1 Clamp main shaft at dog splining.
- 2 Install needle bearing and third speed gear.
- 3 Assemble synchromash member, referring to 7.1/2.
- 4 Press synchromash member in position.



- 5 Install needle bearing, forth speed gear and bearing bush.

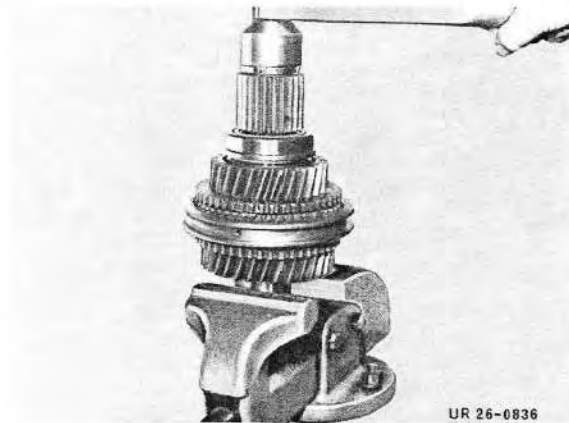
717.9
SA 35 977

6 Press taper roller bearing into position.

7 Press sun gear into position.

Note: Oil drillings must be aligned.

8 Using special tool No. 8, torque sun gear and then secure by knocking with chisel.
See 1.7/1 for tightening torques.



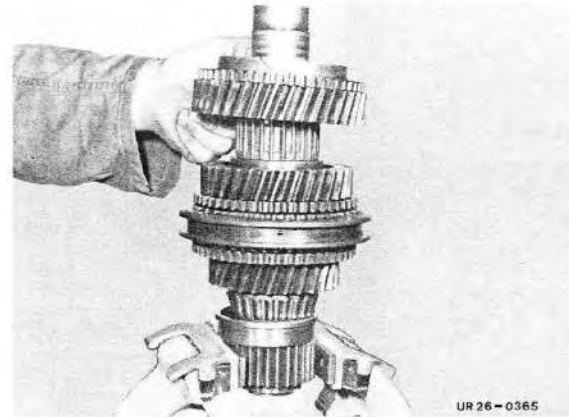
UR 26-0836

9 Turn main shaft by 180°.

10 Install needle bearing and second speed gear.

11 Assemble synchromash member, referring to 7.1/2.

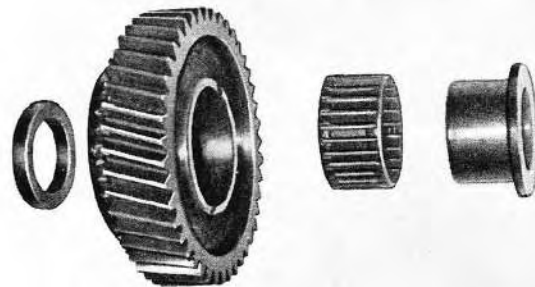
12 Press synchromash member on.



UR 26-0365

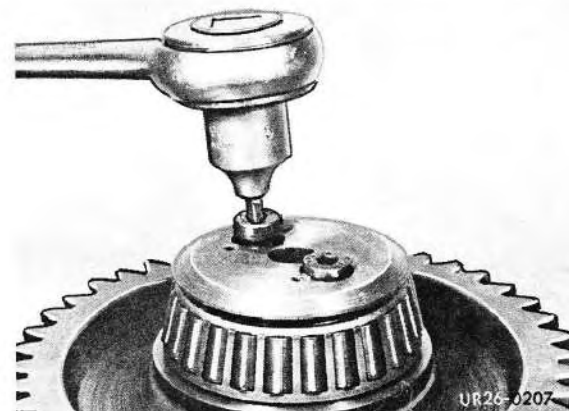
13 Press spacing ring, first speed gear, needle bearing and bearing bush onto shaft.

14 Press inner race of taper roller bearing on.



UR 26-0238

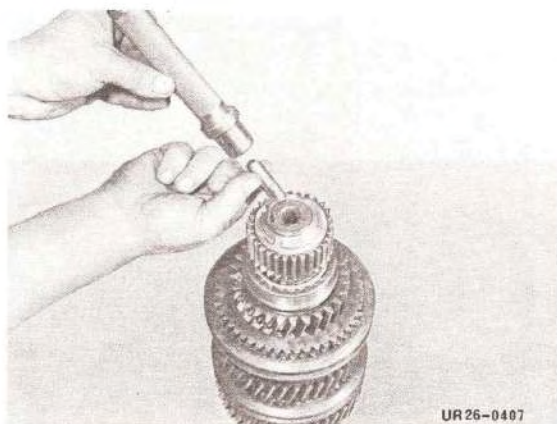
15 Attach supporting plate, torque fastening bolts and secure using center punch.
See 1.7/1 for tightening torques.



UR 26-0207

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16 Using special tool No. 10, press spacing sleeve into position.

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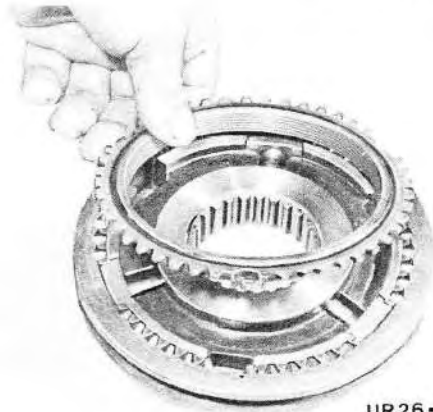
Disassembly

- 1 Remove synchronizing cone.



UR26-0396

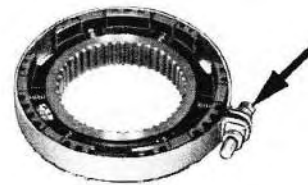
- 2 Remove sliding sleeve.
- 3 Remove driver, pin and spring.



UR26-0396

Assembly

- 1 Hold spring, pin and driver with clip and install sliding sleeve.
- 2 Install synchronizing cone.



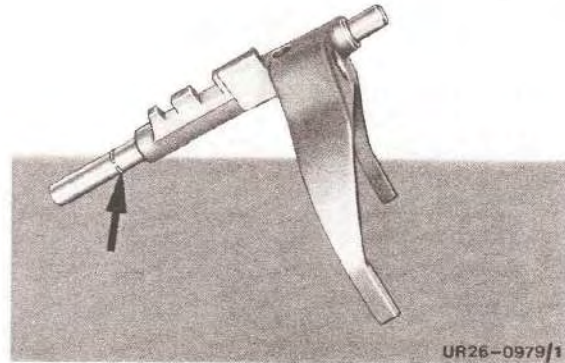
UR26-0981/1

Disassembly

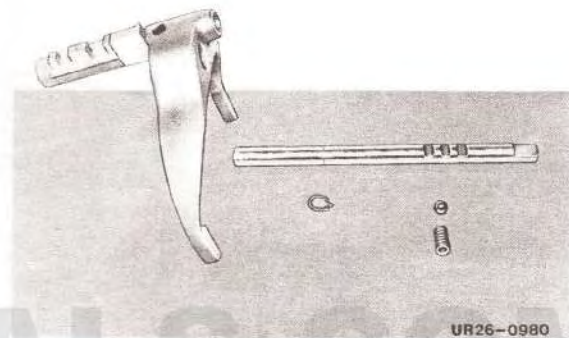
- 1 Disassemble gearbox, referring to 3.5/1.
- 2 Remove snap ring from shift shaft.
- 3 Remove shift shaft.

Danger:

Lock is secured with spring and ball bearing.

**Assembly**

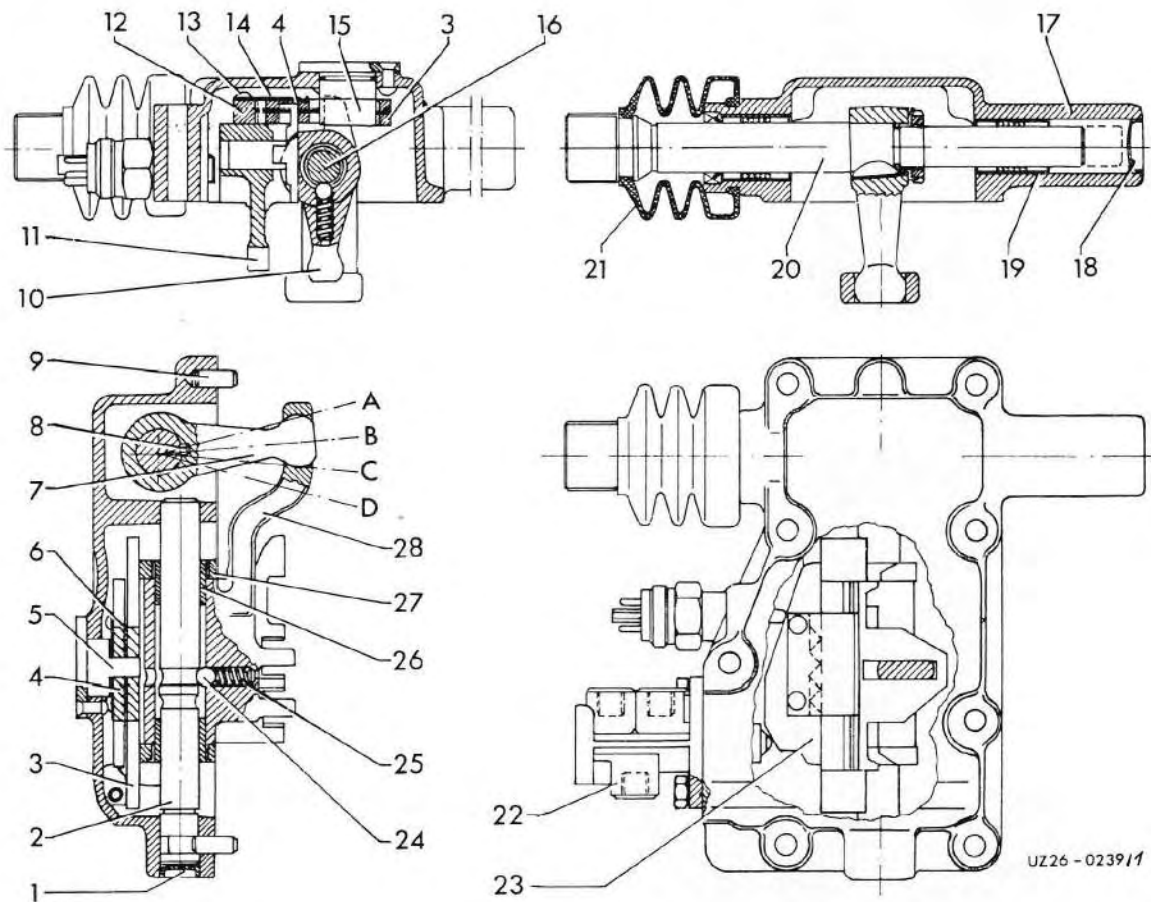
- 1 Install and secure spring and ball bearing in shift shaft.
- 2 Install shift shaft in selector fork.
- 3 Fit snap ring on shift shaft.
- 4 Assemble gearbox, referring to 4.1/1.



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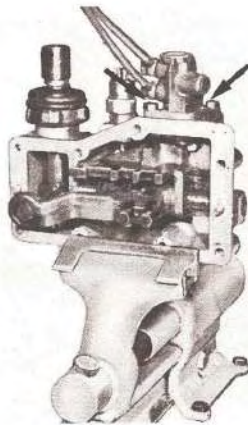
Sectional view



Shift cover

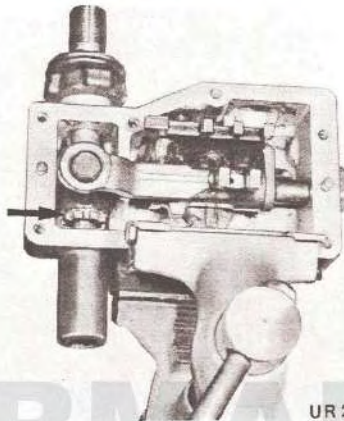
- | | | |
|-----------------------|--|----------------|
| 1 End cover | 15 Guide flange | |
| 2 Shift shaft | 16 Shift shaft | |
| 3 Shift locking plate | 17 Shift cover housing | |
| 4 Shift locking plate | 18 End cover | |
| 5 Guide flange | 19 Bushing | |
| 6 Separating plate | 20 Selector/shift rod | |
| 7 Shift finger | 21 Gaiter | |
| 8 Woodruff key | 22 4/2 directional valve for range shift | |
| 9 Straight pin | 23 Shift lock | |
| 10 Shift finger | 24 Ball, 8 mm | |
| 11 Tappet | 25 Compression spring | A 1st/2nd gear |
| 12 Control cam | 26 Bushing | B 3rd/4th gear |
| 13 Grooved stud | 27 Shift lock | C 5th/6th gear |
| 14 Locking spring | 28 Shift finger | D 7th/8th gear |

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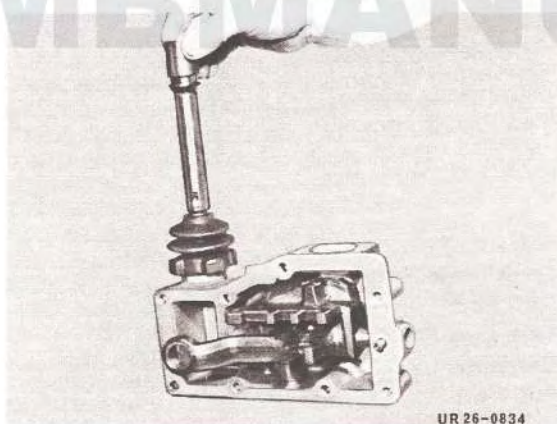


Disassembly

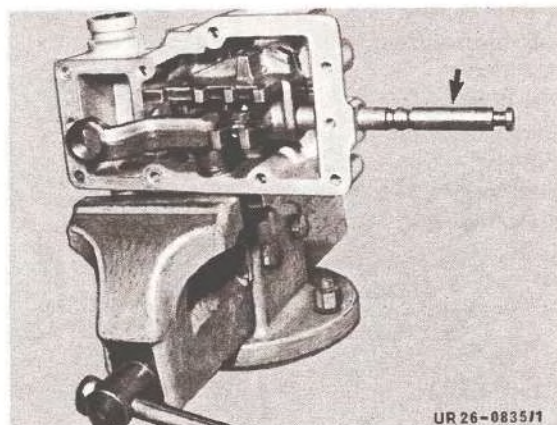
- 1 Clamp shift cover.
- 2 Unscrew shift valve for planetary gearset.
- 3 Unscrew checking switch for shifts.
- 4 Unscrew slotted nut in front of shift finger.



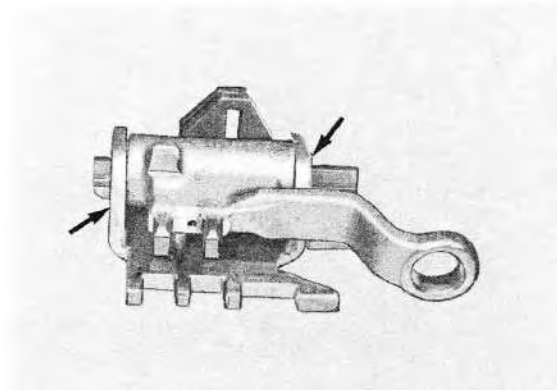
- 5 Remove selector/shift rod.



- 6 Remove dowel pins and shift shaft.
- 7 Remove shift finger.

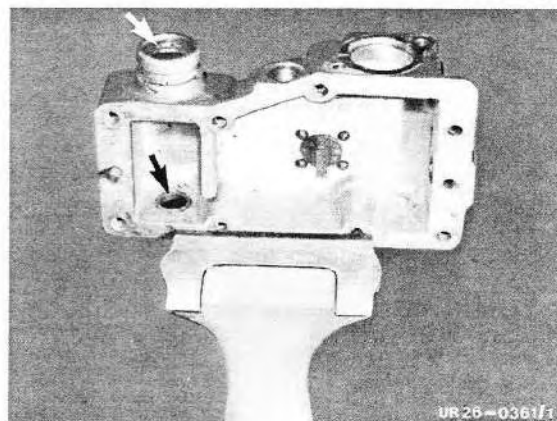


- 8 Remove bushings in shift finger.



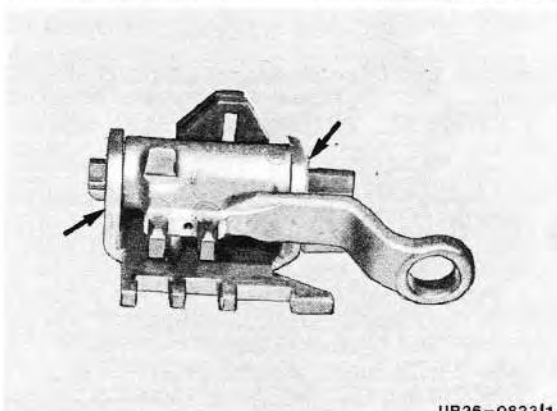
- 9 Remove bushings for selector/shift rod.

- 10 Clean and check all parts, exchanging if necessary.

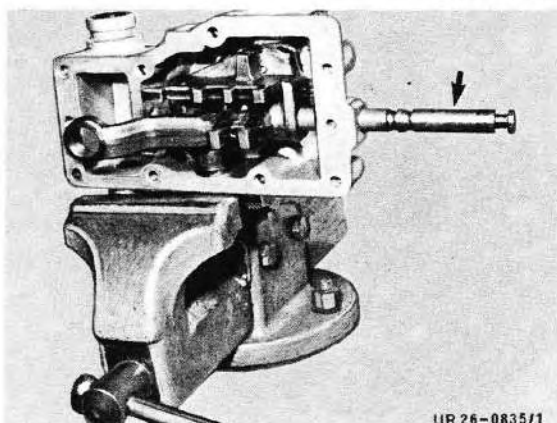


Assembly

- 1 Install bushings for selector/shift rod.
- 2 Install bushings in shift finger.
- 3 Install shift finger.

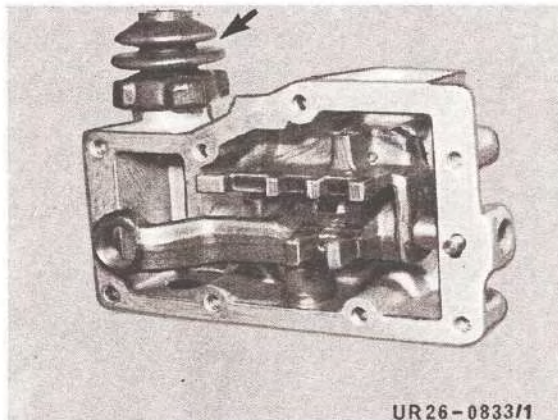


- 4 Install shift shaft.
- 5 Coat shaft seal for shift rod with lubricating grease No. 3 and install in shift cover.



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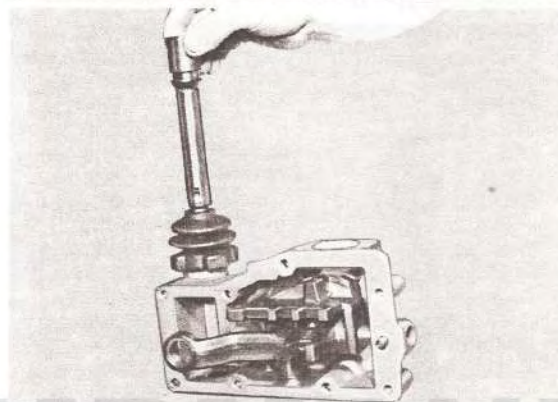
SA 35 977



UR 26-0833/1

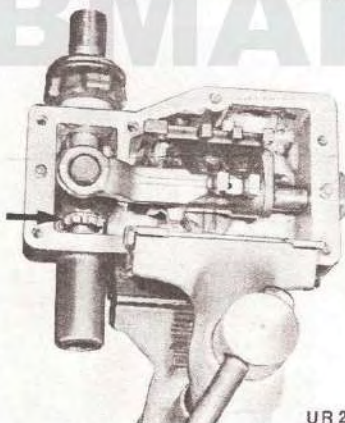
Insert end cover with locking fluid No. 1.

Install and secure gaiter at selector/shift rod.



UR 26-0834

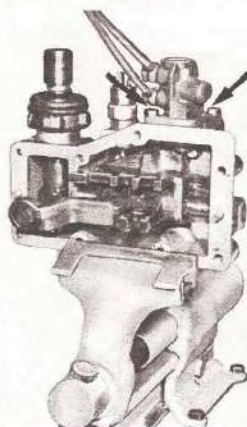
Install shift rod with key.



UR 26-0820/1

Tighten and secure slotted nut in front of shift finger.

10 Install checking switch for shifts.

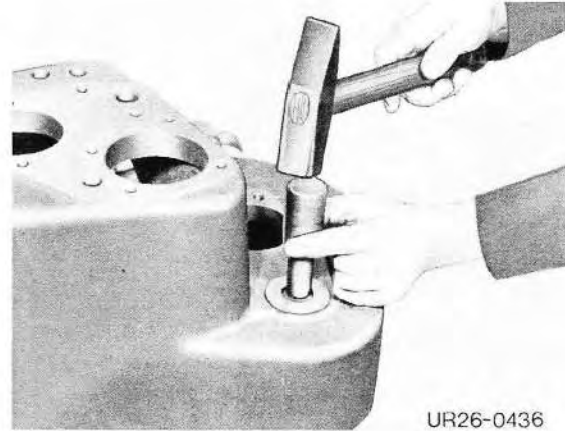


UR 26-0819/1

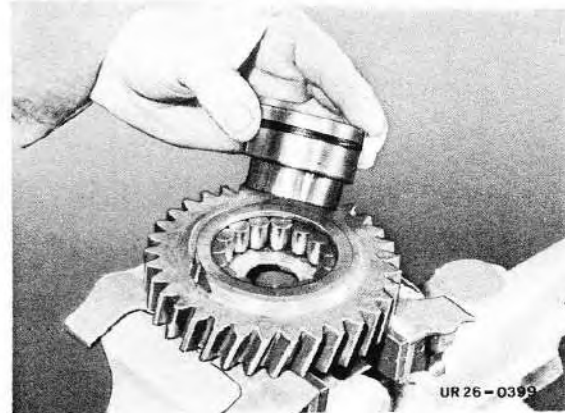
11 Attach shift valve for planetary gearset.

717.901
SA 35 977**Removal**

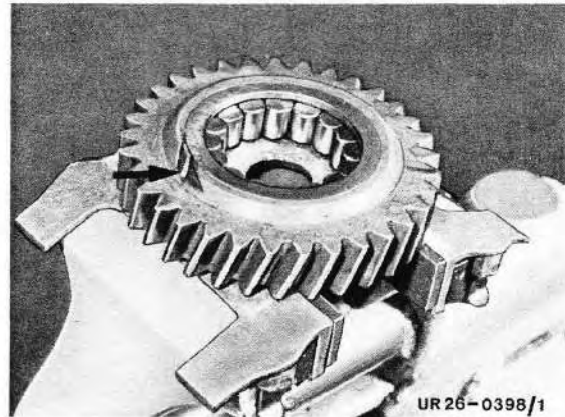
- 1 Force bearing shaft out.



- 2 Remove bearing bush.

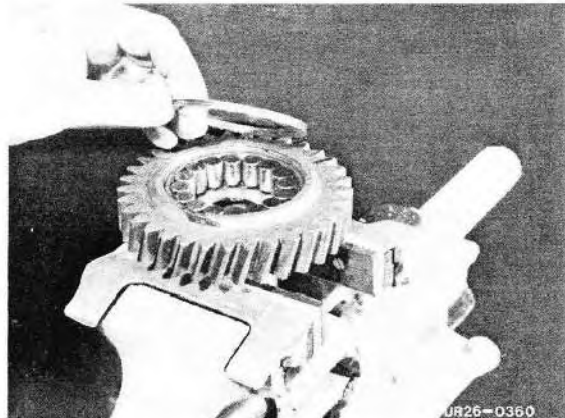


- 3 Remove wire retainer.

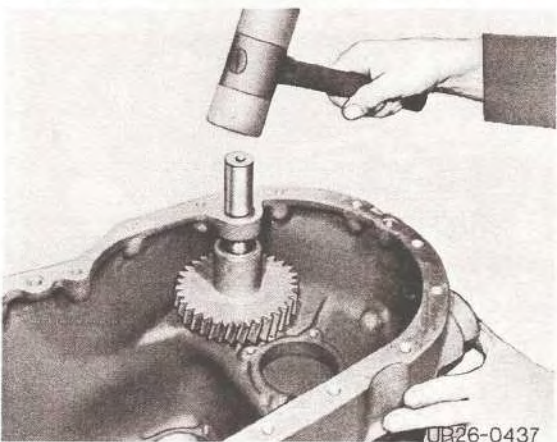
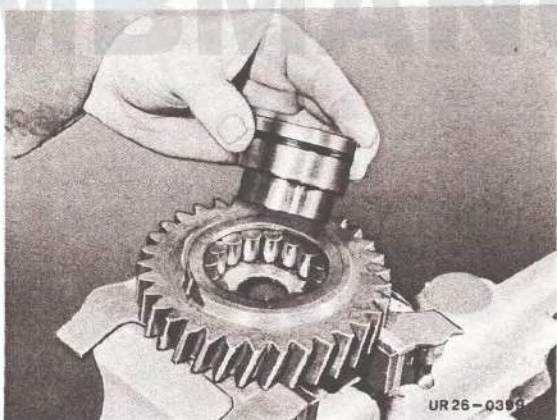
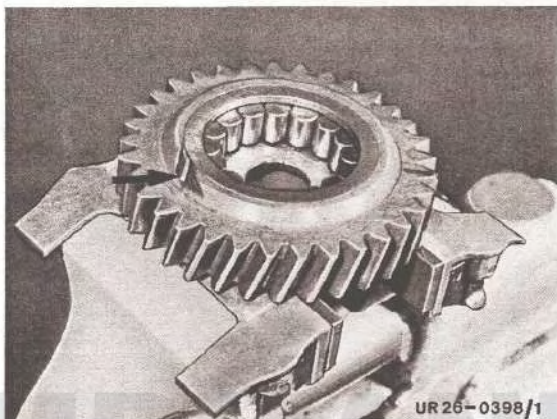
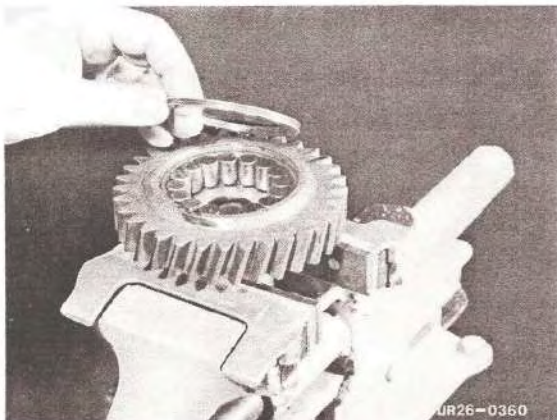


- 4 Remove thrust washers and bearing rollers.

- 5 Take needle bearing out of shift housing.



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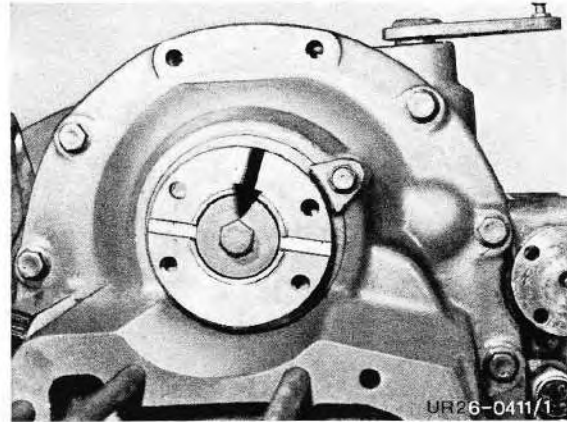


Installation

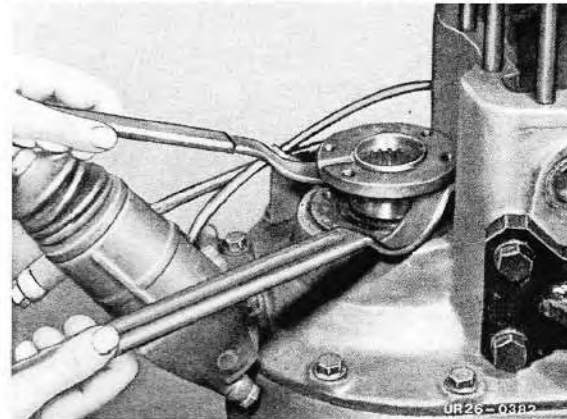
- 1 Install needle bearing in shift housing.
- 2 Insert and fit small thrust washer, bearing rollers and large thrust washer with lubricating grease No. 3.
- 3 Knock wire retainer in.
- 4 Fit bearing bush.
- 5 Fit bearing shaft.

Removal

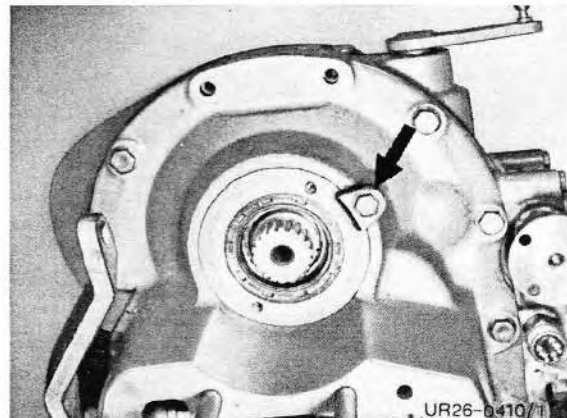
1 Screw fastening bolt out of drive flange.



2 Force drive flange off.

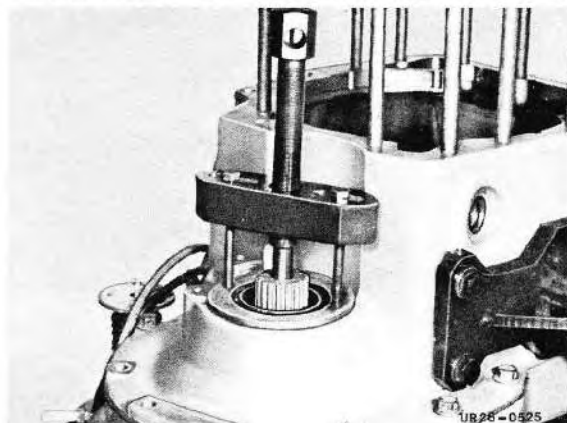


3 Remove oil pump fastening bolt.



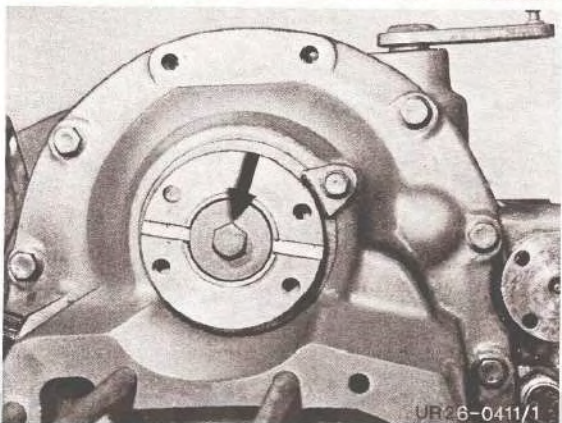
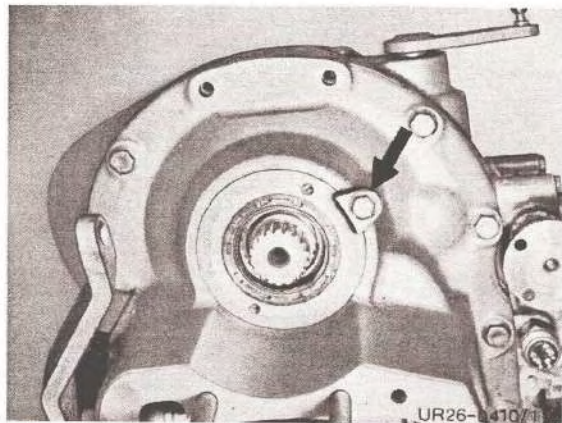
4 Detach oil pump using special tool No. 2.

5 Remove key.



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Installation

- 1 Install oil pump with key and tighten.
See 1.8/1 for tightening torques.

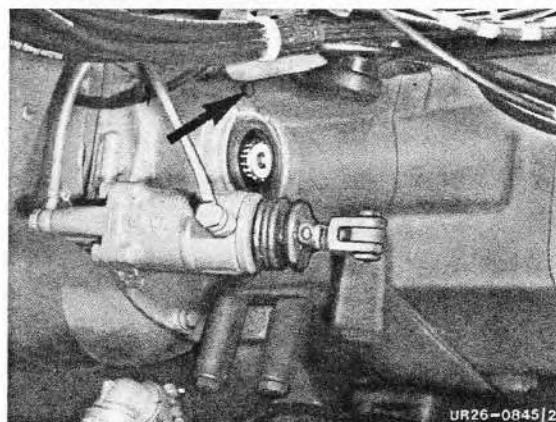
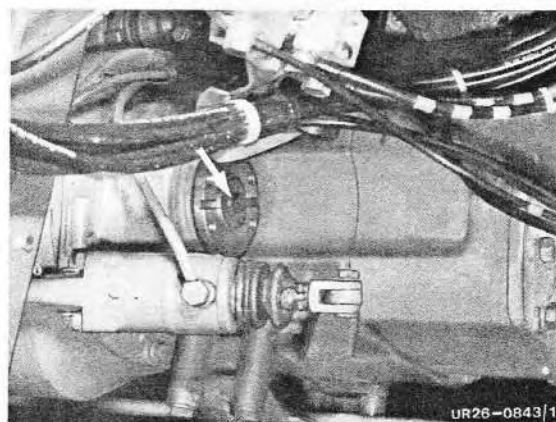
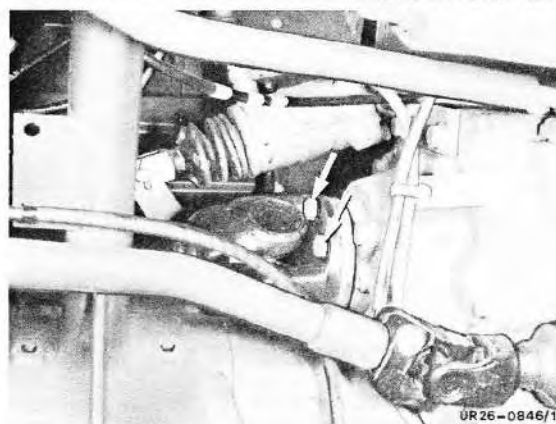
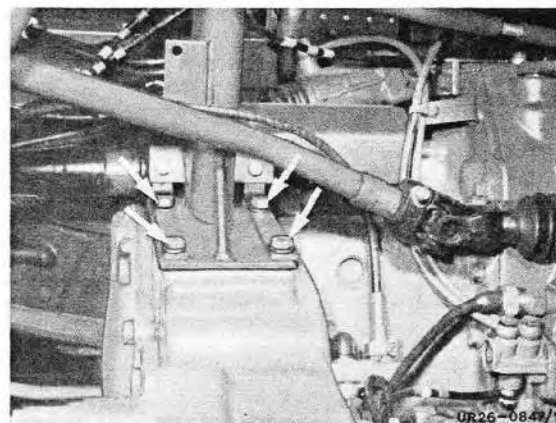
Note: From transmission No. 004 895 a 5 mm wide key is fitted. The previous version cannot be replaced by the new version of the oil pump.

From transmission No. 008 049 a spring lock washer is positioned below the oil pump fastening bolt.

- 2 Press drive flange into position and tighten supporting disk.
See 1.7/1 for tightening torques.

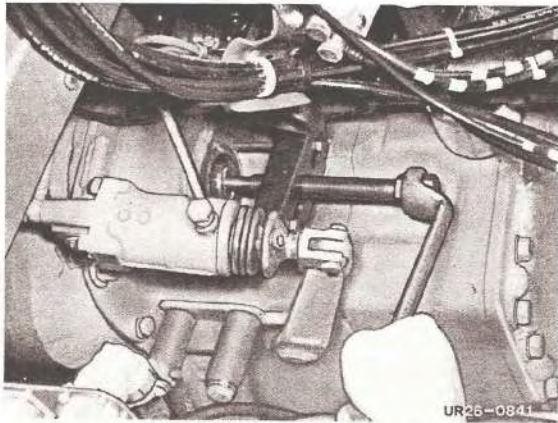
Removal, main transmission installed

- 1 Raise cab, referring to 60.5-2.1/1 or 60.5-2.2/1.
- 2 Detach metal cover.
- 3 Detach input shaft.
- 4 Unscrew fastening bolt and force drive flange off.
- 5 Unscrew oil pump fastening bolt.

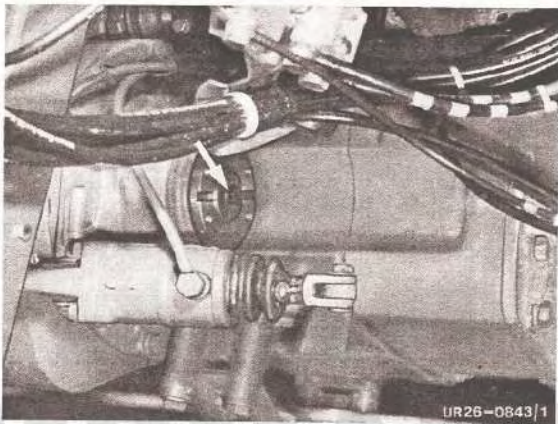


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- 6 Attach special tool No. 2 and remove oil pump.

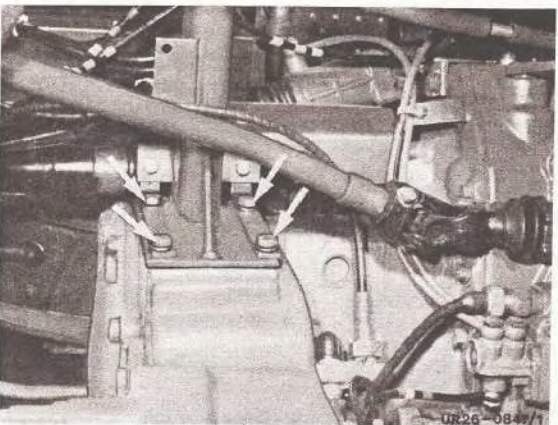


Installation, main transmission installed

- 1 Fit oil pump with key.
- 2 Insert oil pump fastening bolt and torque. See 1.7/1 for tightening torque.
- 3 Press drive flange into position. Torque supporting disk and sealing ring. See 1.7/1 for tightening torques.



- 4 Attach input shaft.

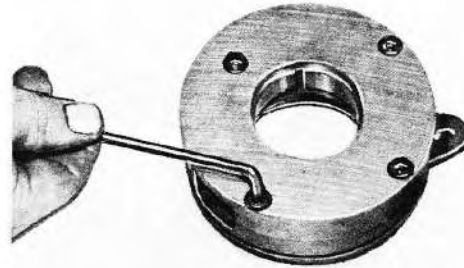


- 5 Attach safety support and metal cover. Coat bolts with sealant No. 1 prior to insertion.
- 6 Lower cab, referring to 60.5-2.1/1 or 60.5-2.2/1.

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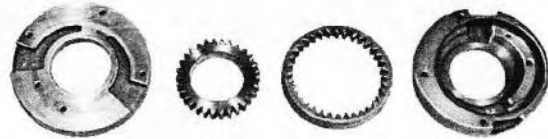
Disassembly

- 1 Remove socket-head screws from oil pump.



UR 26-0383

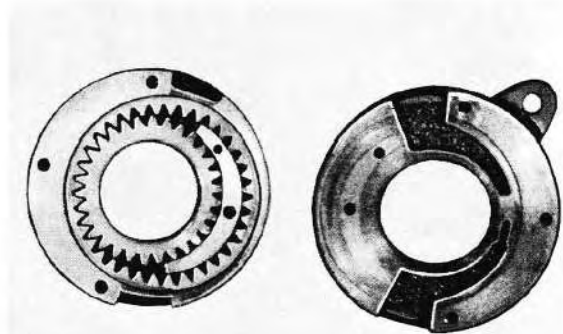
- 2 Disassemble oil pump.
- 3 Remove sealing ring and shaft seal.
- 4 Clean and check all parts, exchanging if necessary.



UR 26-0384

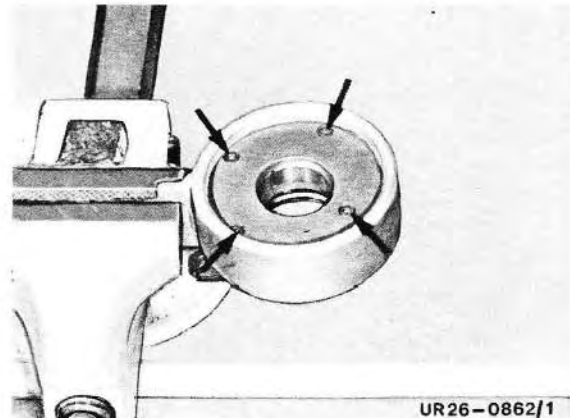
Assembly

- 1 Place both gears in pump body.



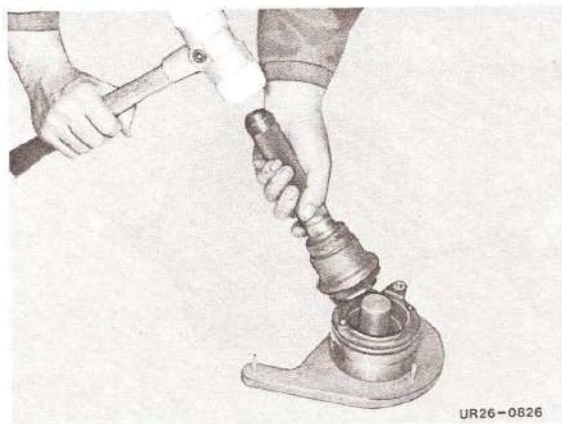
UR 26-0240

- 2 Close two pump body parts together, place in special tool No. 12 and screw together.
- 3 Screw pump housing together.



UR 26-0862/1

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- 4 Coat shaft seal with lubricating grease No. 3 and press into position using special tool No. 1.

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